Lecture - 1

CO4: Compare various statistical methods of study data samples

CO5: Analyze and evaluation of different sets of data using hypothesis testing.

INTRODUCTION

You have studied one-dimensional random variables and their probability mass functions, density functions and distribution functions. There may also be situations where we have to study two-dimensional random variables in connection with a random experiment. For example, we may be interested in recording the number of boys and girls born in a hospital on a particular day. Here, ‘the number of boys’ and ‘the number of girls’ are random variables taking the values 0, 1, 2, … and both these random variables are discrete also.

Now, we concentrate on the two-dimensional discrete random variables defining them. The joint, marginal and conditional probability mass functions of two-dimensional random variable are described in this Unit. The distribution function and the marginal distribution function are discussed in this Unit.

Objectives

A study of this unit would enable you to:

- define two-dimensional discrete random variable;
- specify the joint probability mass function of two discrete random variables;
- obtain the marginal and conditional distributions for two-dimensional discrete random variable;
- define two-dimensional distribution function;
- define the marginal distribution functions; and
- solve various practical problems on bivariate discrete random variables.